UNIVERSITY OF BIRMINGHAM

Year 4 Theory Project Report

Driven Vortex Motion in a Single Channel modelled using a Continuous Potential

Nathan Mayes
1529080

supervised by
Prof. Nicola Wilkin and Dr. Richard Mason

Abstract

We will present simulations and results of the behaviour of vortex matter in a type II superconductor. Using Langevin dynamics, we simulate vortex matter in a homogeneous magnetic field. Firstly, through comparison to work by Spencer and Jensen (Spencer, S. & Jensen, H.J. *Phys. Rev. B*, **55**(13), 8473–8481, 1997), we verified our choice of approximation of the Bessel function that governs vortex-vortex interactions by melting the triangular lattice and comparing the measured values of T_m. A force approximation with a cutoff range $r_{cut} = 3\lambda$ is chosen as the Bessel function that normally describes the forces is slow to compute. Next, we constructed the system of flow in a single commensurate channel using the same approximation for the force. We measured a value of the critical shear force $F_c = 0.055$, and through comparison to the value measured by Besseling (Besseling *et al. New Journal of Physics*, 7:71–71, 2005), we confirmed the validity of our simulations.

We then present the result of investigation into finding an analytic form of the potential, $V(x, y)$, in a single channel due to a pinned channel edge, given by

$$V(x,y) = \frac{\pi}{a_0} \sum_{n=-\infty}^{\infty} \frac{e^{\frac{i2n\pi x}{a_0} + \frac{Q_n}{2}(b_0 - L)}}{Q_n \sinh\left(\frac{b_0 Q_n}{2}\right)} \cosh\left(\left(y - \frac{L}{2}\right) Q_n\right) \left(1 + (-1)^n e^{-\frac{b_0 Q_n}{2}}\right)$$

with

$$Q_n \equiv \sqrt{\frac{1}{\lambda^2} + \left(\frac{2n\pi}{a_0}\right)^2}$$

This representation was constructed using the integral form of the Bessel function and by applying a forward and back Fourier transform. Making use of various identities, we were able to express the potential in terms of elementary functions over a single sum, rather than the square sum required when using the Bessel function. By differentiating with respect to x and y, we obtained an expression for the components of force in the channel, also expressed in terms of elementary functions, allowing for a significant speed up of our simulation.

We analysed the convergence of this sums, and found that on $y = L/2$ the value of the potential converges to within $\approx 3 \times 10^{-6}$ of the true value of the Bessel function (calculated considering vortices at a distance of 100 lattice spacings in both directions) for $|n| \leq 1$. We also found that the y-component of force has a zero at $y = L/2 = b_0/2$ for our single channel commensurate system, and so by constructing our system to have vortices initially at $y = b_0/2$, we ensure that they remain on $y = b_0/2$, and as such we use the $|n| \leq 1$ approximation of the potential in our simulation.

The simulation of flow of vortex matter due to an external driving force was then repeated, using the analytic form to describe the channel edges, and measured a new value of the critical shear force $F_c = 0.05$. This discrepancy is because the discrete case requires a larger value of r_{cut} to calculate the true value of the force on a vortex, and as such we expect that this new value of F_c is the correct value.

Lastly, we discussed the different possible ways we might wish to alter the geometry of our system, and how this would be easier to implement using our analytic form potential than it would be using Bessel functions, allowing for a new avenue into analysing channel flow systems, with drastically reduced simulation times.

Contents

1 Motivation **3**

2 Background Physics **5**
2.1 History of Superconductors . . . 5
2.1.1 Fundamental Properties . . . 5
2.1.2 The London Equations . . . 5
2.1.3 Ginzburg-Landau Theory . . . 6
2.1.4 BCS Theory . . . 7
2.2 Type II Superconductors . . . 7
2.2.1 Flux Quantisation . . . 7
2.2.2 Abrikosov Lattice . . . 7
2.2.3 Commensurability . . . 8
2.2.4 Force Between Flux Lines . . . 8
2.2.5 Flux Line Motion . . . 9
2.2.6 Lattice Melting . . . 10
2.3 Crystal Defects . . . 10
2.3.1 Motivation . . . 10
2.3.2 Point Defects . . . 11
2.3.3 Dislocations . . . 11
2.3.4 Grain Boundaries . . . 11

3 Modelling **13**
3.1 Lattice Geometry . . . 13
3.2 Lattice Dynamics . . . 13
3.2.1 Motivation . . . 13
3.2.2 Stochastic Term . . . 13
3.2.3 Vortex-Vortex Interaction . . . 14
3.2.4 External Forces . . . 15
3.2.5 Numerical Method . . . 15

4 Analysing the Simulation **16**
4.1 Basic Tests . . . 16
4.1.1 Two Particle State . . . 16
4.1.2 Hexatic State . . . 16
4.2 Analysis Techniques . . . 16
4.2.1 Hexatic Order Parameter . . . 16
4.2.2 Density of Defects . . . 16
4.2.3 Lattice Melting . . . 17
4.2.4 Critical Shear Force . . . 18

5 Derivation of Analytic Form Boundary Potential **20**
5.1 Motivation . . . 20
5.2 Infinite Square Lattice . . . 20
5.3 Semi Infinite Square Lattice . . . 21
5.4 Extension to Channel System . . . 22
5.5 Forces due to Analytic Potential . . . 22
5.5.1 F_x . . . 22
5.5.2 F_y . . . 22
5.5.3 Channel Vortex Interactions . . . 23

6 Verification of the Analytic Potential **24**
6.1 Convergence . . . 24
6.1.1 Convergence along x-axis . . . 24
6.1.2 Convergence along y-axis . . . 25
6.1.3 Conclusions . . . 26
6.2 Critical Shear Force . . . 27
6.2.1 Results . . . 27
6.3 Computation Times . . . 27

7 Future Work **29**
7.1 Modulation of the Analytic Potential . . . 29
7.2 Altering Lattice Geometry . . . 29

8 Conclusion **30**

1 Motivation

The topics discussed in this report relate to the simulation and analysis of vortex matter in type II superconductors. In these superconductors magnetic flux is quantised when the applied field is between critical values H_{c1} and H_{c2} and penetrates the material in isolated cores contained by supercurrent vortices. These vortices arrange themselves in a triangular lattice at low temperatures, but due to thermal fluctuations amongst other reasons, the lattice structure can be disrupted and dislocations can form, i.e. regions where the 6-fold symmetry of the triangular lattice becomes either 5- or 7-fold. For large enough thermal fluctuations, the vortex matter can undergo a phase transition to a liquid phase, causing a loss of long range translational and rotational order, with the number of dislocations increasing rapidly at the transition.

One system that is of interest to us is the thin channel superconductor, which is formed by taking a double layer of superconducting material and forming a channel of width w and length L. We can see an example of such a superconductor in figure 1.1. The upper layer consists of a material with defects such that when a magnetic field is applied at low temperature, the vortices become pinned at the edges of the channel, whilst the lower layer is weakly pinned, and therefore the vortices in the channel are free to move. As magnetic flux is quantised, we can calculate the number of vortices from the magnetic field strength.

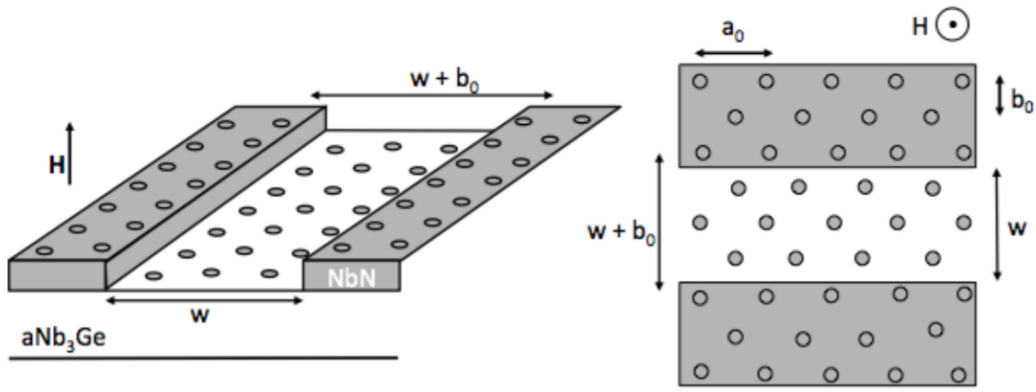

Figure 1.1: An example of a narrow channel superconductor. The top layer is made from NbN and by removing a section of width w it reveals the channel of aNb_3Ge that is weakly pinned. Figure from Ref[1]

The motivation for this project comes from the work of a few groups [1, 2, 3] who have analysed various properties of the thin channel system. However the aim of this project is not the analysis and variation of key parameters of the thin channel system, but to attempt to find a new way to model our simulations. We will attempt to do this by using a continuous analytic form of the potential in the channel to describe the channel edges, instead modelling it as a discrete set of pinned vortices. The reasoning for this is that when calculating the force on a free vortex, we must consider the force due to all vortices within a cutoff

range. The force between two vortices is proportional to a Bessel function, and for large enough systems, the number of calculations of this Bessel function is very large, and therefore our system can take a long time to simulate. If we can replace the boundary by a continuous function, all contributions to the force on a vortex due to the vortices in the boundary can be reduced to one calculation. As well as this, we can apply modulations to the boundary potential, giving a much simpler method to investigate the behaviour of the system. Instead of rewriting our code to increase or decrease the density of vortices in the boundary, we are only required to differentiate the new potential and implement the new expressions of the forces.

2 Background Physics

2.1 History of Superconductors

2.1.1 Fundamental Properties

The phenomena of superconductivity was first discovered by H.K. Onnes [4] in his experiments cooling pure metals with liquid helium, allowing him to reach temperatures of just a few degrees above absolute zero. He found that below a transition temperature T_c, dependent on the material, the metal acted as a perfect conductor, which he called *perfect superconductivity*, meaning that zero electrical resistance was measured.

The next fundamental property of superconductors was discovered in 1933 by Meissner and Ochsenfeld [5], where they demonstrated *perfect diamagnetism*. They found that below T_c, there was complete flux expulsion inside the superconducting material except within a small penetration depth λ. This can be explained by the formation of "supercurrent" on the surface of the material due to an applied magnetic field. These supercurrents induce a magnetisation in the material and below a critical applied field, H_c, for pure metal superconductors, the applied field cancels with the induced field, with no flux left to penetrate, except within the penetration depth λ. However above H_c the superconducting state is lost as flux begins to penetrate the material.

The localised field penetrating the material is given by the sum of the applied field, $\mathbf{H}$, and the magnetisation caused by the surface supercurrents, $\mathbf{M}$. Hence the local field is $\mathbf{B} = \mu_0\mathbf{H} + \mathbf{M}$ and in the superconducting phase we have $\mu_0\mathbf{H} = -\mathbf{M}$.

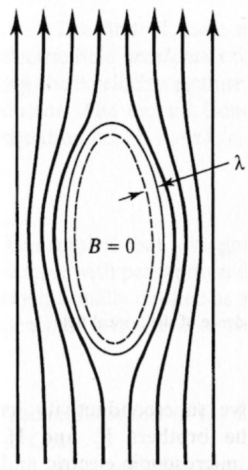

Figure 2.1: A visualisation of flux expulsion in a type I superconductor. Figure from Ref[6]

2.1.2 The London Equations

The work done by Meissner and Ochsenfeld on flux expulsion quickly led to further developments in 1935 by the London brothers[7]. They developed a set of equations relating electric current and magnetic field inside a superconductor. They started from a perfect conductor consisting of a gas of superconducting charge carriers with number density n_s, charge e^*, and moving without collision at velocity $\mathbf{v}$, giving a current of $\mathbf{J_s} = n_s e^* \mathbf{v}$. From Newton's 2nd Law in an electric field

$$m^*\dot{\mathbf{v}} = -e^*\mathbf{E} \implies \dot{\mathbf{J_s}} = \frac{n_s e^{*2}}{m^*}\mathbf{E} \tag{2.1}$$

Using the Maxwell equation $\nabla \wedge \mathbf{E} = -\dot{\mathbf{B}}$ they showed:

$$\nabla \wedge \dot{\mathbf{J_s}} = -\frac{n_s e^{*2}}{m^*}\dot{\mathbf{B}} \tag{2.2}$$

Then assuming that $\mathbf{B}$ is not rapidly changing with time, which allowed them to ignore the term generated including its third time derivative, this gave

$$\nabla^2\dot{\mathbf{B}} = \frac{1}{\lambda^2}\dot{\mathbf{B}} \tag{2.3}$$

which describes a field that penetrates a superconducting material in the normal state, which becomes locked inside the material once cooled to the superconducting state. However as Meissner[5] had already discovered this was not what was observed experimentally, as flux was expelled in all cases, so they attempted to alter their ideas. They did this by removing the time dependence from equation 2.2, giving them

$$\dot{\mathbf{J}}_\mathbf{s} = \frac{n_s e^{*2}}{m^*}\mathbf{E} \quad (2.4) \qquad \nabla \wedge \mathbf{J}_\mathbf{s} = -\frac{n_s e^{*2}}{m^*}\mathbf{B} \quad (2.5) \qquad \nabla \wedge \mathbf{B} = \mu_0[\mathbf{J} + \epsilon_0 \dot{\mathbf{E}}] \quad (2.6)$$

and following the same process as before they found

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda^2}\mathbf{B} \qquad \text{with } \frac{1}{\lambda^2} = \frac{\mu_0 n_s e^{*2}}{m^*} \tag{2.7}$$

The London equations are important as they provide a description of the superconducting state in terms of the electric and magnetic fields, as well as introducing two important parameters, n_s, the density of superconducting electrons, and λ, the magnetic penetration depth.

2.1.3 Ginzburg-Landau Theory

In 1950 Ginzburg and Landau (GL)[8] developed another theory to explain the superconducting state. Based on Landau's earlier work, they said that the free energy F of a superconductor near the transition to the superconducting state can be expressed as a function of a complex order parameter, ψ, which is zero in the normal state and nonzero below the phase transition into the superconducting state. It is related to the density of superconducting electrons by $n_s = |\psi(x)|^2$ and this order parameter is analogous to the wavefuction describing the superconducting electrons. GL theory also predicts a new length scale, $\xi(T)$, known as the coherence length, which is the scale at which the magnetic field varies and the GL parameter is defined as $\kappa = \lambda/\xi$, where λ is the London penetration depth. In the previous work done by Meissner and Ochsenfeld, and Onnes with conventional superconductors, we had $\kappa \ll 1$, which is described by a positive energy cost between normal and superconducting domains.

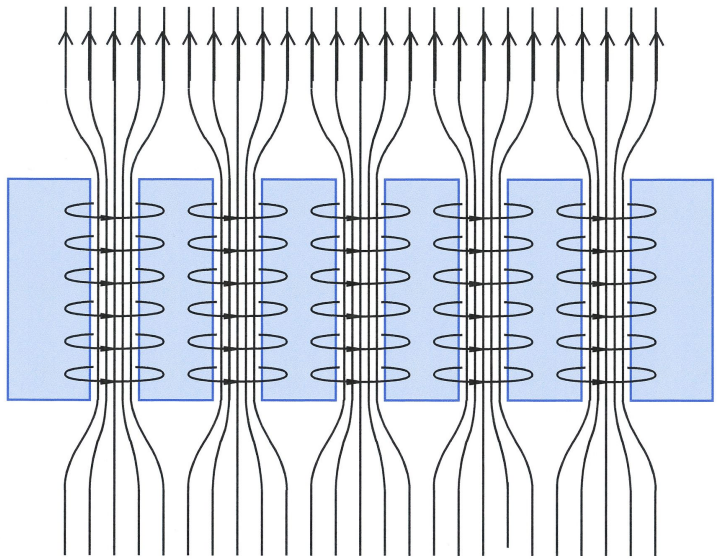

Figure 2.2: Cores of magnetic flux through a type II superconductor, separated by supercurrent vortices. Figure from Ref[9]

Further work was done in 1957 by Abrikosov[10], one of Landau's students, when he considered the $\kappa \gg 1$ case, i.e. when the coherence length is greater than the penetration depth. He found there is a negative energy between the interfaces of the superconducting and normal phases, causing multiple domain walls to form, and the surface area to be maximised. This is manifested as cores of magnetic flux penetrating the material, with the normal region separated from the superconducting domain by supercurrent vortices as seen in figure 2.2. He named this state type II superconductivity, and found the

transition between type I and type II happened at $\kappa = 1/\sqrt{2}$. We see that for an applied field H, there is an upper, H_{c2}, and lower, H_{c1}, critical field. At H_{c2} we start to see flux penetration before it goes fully superconducting at H_{c1}. We call the critical field for a type I superconductor H_c and $H_{c1} < H_c < H_{c2}$[6].

2.1.4 BCS Theory

In 1957, Bardeen, Cooper, and Schrieffer (BCS) developed a mathematically complex theory to explain the microscopic origins of superconductivity[11]. In a superconducting material, an electron causes a distortion in the cation lattice around it, increasing the positive charge density in this region. This charge distortion, or *phonon*, attracts another electron some distance away and causes the electrons to become bound in a *Cooper pair* due to this electron-phonon interaction. Cooper pairs are the carriers of the supercurrent, and resist vibrations in the lattice due to the attraction to the other electron. This mechanism explains perfect conductivity, as below T_c the pairs can move in the lattice mostly unaffected by thermal vibrations. BCS theory fails at high temperature, as it predicts that T_c has a maximum value of $30 - 40K$, and in 2015, superconducting behaviour was observed at high pressure, with a T_c of $203K$[12].

2.2 Type II Superconductors

2.2.1 Flux Quantisation

Each normal region or hole passing through the superconductor has an associated fluxoid, Φ'. London[6] gave the definition of this to be

$$\Phi' = \Phi + \left(\frac{4\pi}{c}\right) \oint \lambda^2 \mathbf{J}_s . d\mathbf{s} = \Phi + \left(\frac{m^* c}{e^*}\right) \oint \mathbf{v}_s . d\mathbf{s} \tag{2.8}$$

where $\Phi = \oint \mathbf{A}.d\mathbf{s}$ is the magnetic flux through the integration circuit. He used the Bohr-Summerfeld quantisation condition to show

$$\oint \mathbf{p}.d\mathbf{q} = nh \tag{2.9}$$

$$\implies \Phi' = \frac{c}{e^*} \oint \left(m^* \mathbf{v}_s + \frac{e^* \mathbf{A}}{c}\right) .d\mathbf{s} = \frac{c}{e^*} \oint \mathbf{p}.d\mathbf{s} = nh\frac{c}{e^*} = n\Phi_0 \tag{2.10}$$

However as the quasi-particles for superconductivity are Cooper pairs of electrons we actually have $e^* = 2e$ and $m^* = 2m_e$. We therefore have the flux quantum given as:

$$\Phi_0 = \frac{hc}{2e} \tag{2.11}$$

2.2.2 Abrikosov Lattice

Abrikosov[13] also managed to calculate how the flux lines penetrating the superconductor formed themselves in a stable arrangement. Initially he calculated this to be a square lattice due to a numerical error, but it turned out to be triangular, known as the *Abrikosov Lattice* for a superconductor, or also the flux line lattice (FLL). We know that the flux through a core is quantised, therefore we can calculate the number density of the lattice, as the number of vortices in a certain region is related to the local field in this region B, and the flux quantum Φ_0, by

$$n_v = \frac{B}{\Phi_0} \tag{2.12}$$

Clearly, increasing the applied field will increase the number density of the vortices, and as the vortices have size ξ and separation λ there must be an upper limit, corresponding to $\kappa \sim 1$, otherwise the cores will overlap and there would be full flux penetration, i.e. loss of the superconducting phase. The separation of the vortices depend on the strength of the magnetic field and so also the number density, and from this we calculate the vortex spacing for a given B to be

$$a_0^2 = \frac{2\Phi_0}{\sqrt{3}B} = \frac{2}{\sqrt{3}n_v} \tag{2.13}$$

which determines the magnitude of repulsive force felt between vortices. In a triangular lattice, this vortex spacing a_0 is the horizontal spacing within the rows, and the separation between rows is given by

$$b_0 = \frac{\sqrt{3}a_0}{2} \tag{2.14}$$

2.2.3 Commensurability

One important system we will study is flow through a single channel, created by etching a channel of width w and length L in a double layer of the superconducting material. When the channel width is an integer multiple of b_0, the system is called *commensurate*. In Besseling's[2] work in figure 2.3, we see that whether or not the channel is commensurate has a measurable effect on the *critical shear force*, a property of the system that represents the value of an external driving force at which we begin to see vortices flow. We will use this property later to evaluate the validity of our analytic potential. In our work, we choose to work in a commensurate single channel system, partly as the system is much easier to setup, but more importantly so that we can measure a nonzero value of the critical shear force for our analysis.

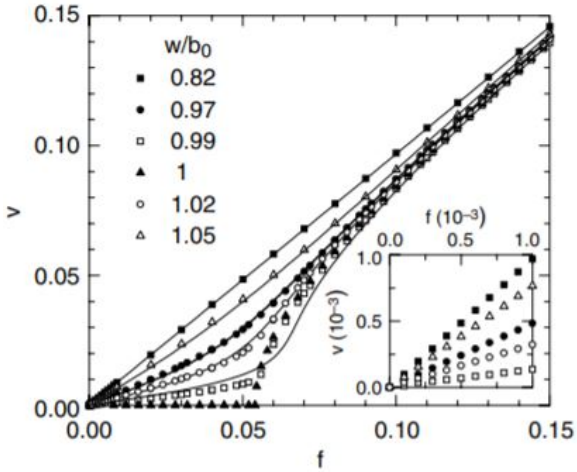

Figure 2.3: For channel widths that are not integer values of the row spacing, we such much different behaviour of the system that for the commensurate case. Figure from Ref[2]

2.2.4 Force Between Flux Lines

Firstly, we deal with the force between flux lines. From the London equations we have the field at a distance r from a vortex core is given by

$$B(r) = \frac{\Phi_0}{2\pi\lambda^2}\int_0^\infty \exp\left(-\frac{r}{\lambda}\cosh t\right)dt = \frac{\Phi_0}{2\pi\lambda^2}K_0\left(\frac{r}{\lambda}\right) \tag{2.15}$$

where $K_0(x)$ is a modified Bessel function of the second kind, with the limits

$$K_0(x) = \begin{cases} -\ln x & x \to 0 \\ \sqrt{\frac{\pi}{2x}}e^{-x} & x \to \infty \end{cases} \tag{2.16}$$

If we work in the limit $\kappa \gg 1$ and $\lambda > a_0$, meaning that the magnetic fields of the vortices are long ranged and overlap with other vortices whilst the cores stay small and isolated as seen in figure 2.4, we can take the linear superposition of the field from two neighbouring vortices[6], and find the interaction energy to be

$$U(r) = \frac{\Phi_0^2}{2\pi\mu_0\lambda^2}K_0\left(\frac{r}{\lambda}\right) \tag{2.17}$$

We can differentiate this equation with respect to position to give an expression for the force ($f(r) = -dU(r)/dr$)

$$f(r) = \frac{\Phi_0^2}{2\pi\mu_0\lambda^3}K_1\left(\frac{r}{\lambda}\right) \tag{2.18}$$

using the relationship $K_0'(r/\lambda) = -K_1(r/\lambda)/\lambda$.

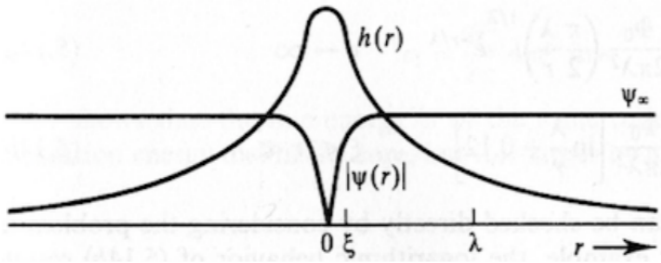

Figure 2.4: For $\kappa \gg 1$, we have that the cores of the vortices are small and isolated, whilst the magnetic fields due to the vortices act over long range and overlap. Figure from Ref[6]

2.2.5 Flux Line Motion

The most common type of motion is caused by an applied current perpendicular to the flux line lattice. This causes a resultant Lorentz (or Magnus) force given by $\mathbf{F}_L = \mathbf{j} \wedge \Phi_0 \mathbf{B}$ which results in a driving force on the vortices. Bardeen and Stephen [14] showed that in a material free from defects the quasi-normal cores interact with the rest of the superconducting material, causing a drag term with viscous drag constant of proportionality

$$\gamma \approx \frac{\Phi_0 H_{c2}}{\rho_n c^2} \tag{2.19}$$

where ρ_n is the resistivity of the material in the normal state. Importantly, γ only depends on H_{c2} and not on B. The flow of flux then induces a voltage in the sample, opposing the direction of flow, and the resistance of this voltage was shown to be $\sim B\rho_n/H_c \neq 0$, making the superconductor lose its zero resistivity property as it dissipates some of the energy from the applied current.

When the material is not free from defects[6], and these defects are of order λ, a vortex may feel a pinning force due to a defect which is strong enough to stop vortex motion, and so the resistive voltage disappears and zero resistivity is retained. These pinning sites cause an energy barrier, but vortices may jump between pinning sites for sufficiently large applied current or thermal activation. For large enough temperatures, bundles of flux may move cooperatively between pinning sites. At nonzero, but low temperature $T < T_m$ there are three distinct regions emerge for $\mathbf{j}$. In the first case:

a) $j = 0$

Figure 2.5: Figure taken from Ref[1]

In this case, flux bundles are confined to local potential wells caused by the pinning sites, as seen in figure 2.5.

b) $0 < j < j_c$

Figure 2.6: Figure taken from Ref[1]

In this case, the flux bundles move cooperatively between pinning sites, seen in figure 2.6, at a slow rate, given by R, which is known as *flux creep*.

c) $j > j_c$

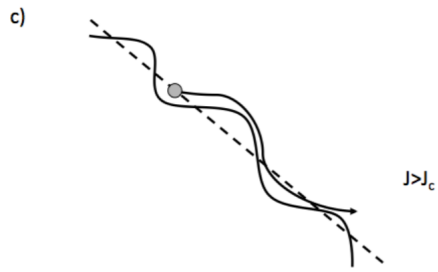

Figure 2.7: Figure taken from Ref[1]

As seen in figure 2.7, when $j > j_c$ there is no energy barrier, and so flux may flow.

2.2.6 Lattice Melting

When temperature is sufficiently high, thermal fluctuations become significant enough that the Abrikosov lattice can melt into a liquid phase of vortices, leading to a loss of long range translational and rotational order above a melting temperature, T_m. T_m can be estimated [6, 15] by assuming that melting in 2D crystalline systems is caused by the nucleation of dislocations. Work by Nelson[15] suggested that in 2D, melting occurs through an intermediate hexatic phase, which is characterised by a short-range translational order and a quasi-long-range rotational order. Beyond this we expect to see a phase characterised fully by short-range translational and rotational order which is what we expect to see in our simulations. Figure 2.8 shows a phase diagram for the thin pancake geometry we work in. We see that the lower critical field H_{c1} drops to zero, removing the possibility of a transition to the Meissner phase[16]. One example of a superconductor that has displayed melting behaviour is $Ba(Fe_{1-x}Co_x)_2As_2$[17]

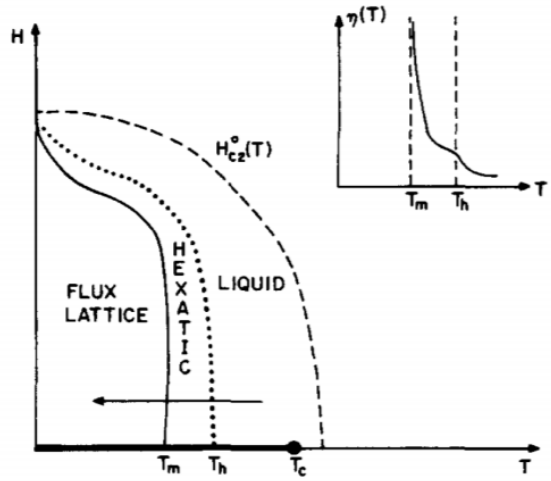

Figure 2.8: The phase diagram for a thin type II superconductor We measure a melting temperature at the transition to the full liquid phase. Figure taken from Ref[16]

2.3 Crystal Defects

2.3.1 Motivation

Crystal defects are significant in many areas of physics, but we will focus on defects in 2D systems. Defects are points or regions where the lattice structure is not perfect, and for the triangular lattice seen in type II superconductors, this corresponds to points with any number of nearest neighbour points other than six.

Defects and disclocations (line defects) are responsible for affecting the physical properties of materials. For example, the way dislocations form and how they move and interact when under stress from an external force determines the malleability of metallic elements. Significantly, as they appear throughout physics, by learning about one system it is possible to shed light on another area of physics.

2.3.2 Point Defects

Point defects are uncommon in the systems we simulate, so are not covered in depth. They occur when a particle is removed, known as a *vacancy*, or added, called an *interstition*. This causes particles close to the defect to be disturbed, causing local changes to the lattice structure.

2.3.3 Dislocations

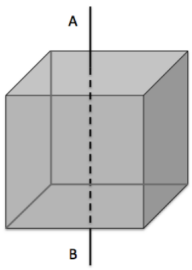

Figure 2.9: The line AB through the system is the line at which the dislocation occurs

Dislocations are line defects and come in one of two types, *edge* and *screw*, or *mixed*. Mixed are made of both edge and screw and are not relevant to our work. To create a screw dislocation we take a perfect lattice seen in figure 2.9 and cut along AB. Then the material on one side of the cut is shifted in the direction parallel to AB. For an edge dislocation we do the same but instead shift the material perpendicular to AB. Figure 2.10 demonstrates edge and screw dislocations. Clearly for a 2D lattice we may only deal with edge dislocations, as we only deal with the plane whose normal is parallel to AB.

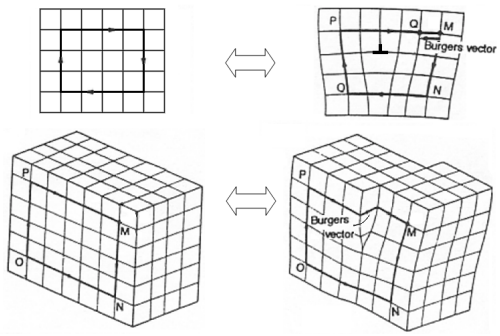

Figure 2.10: An example of an edge and screw dislocation. We can see from this that only edge dislocations are possible in a 2D geometry. Figure taken from Ref[18]

We use the *Burgers vector*, $\mathbf{b}$, to describe these dislocations quantitatively as a measure of the amount of discrepancy in a section of material. It is defined by

$$\mathbf{b} = \oint \frac{\partial \mathbf{u}}{\partial l} dl$$

and for an edge dislocation it is always perpendicular to the dislocation line. We are able to define the glide or slip plane in which a dislocation will move by $\mathbf{b} \wedge \hat{\xi}$ where $\hat{\xi}$ is the direction of the dislocation line.

2.3.4 Grain Boundaries

During crystal formation, materials do not form one perfect crystal, but instead split up into multiple smaller grains. Locally, each grain of material has the same lattice vector orientation, and each grain is

separated by a grain boundary, made from defects and often around a few atoms thick. These boundaries allow the material to be continuous across the grains, and allowing the grains themselves to remain almost stress free. These often complex boundaries can be described by a line of edge dislocations.

3 Modelling

3.1 Lattice Geometry

As in similar work done by other groups, such as Watkins[1] and Besseling[2], we work in a pancake geometry, i.e. a very thin superconductor. Because of this, we can assume that the flux lines penetrating the superconductor are straight all the way through the superconductor. We can then consider the system from the plane perpendicular to the direction of the flux lines, and model the vortices as a system of particles in a 2D plane. For simplification, in our triangular lattice we choose $a_0 = 1$ and hence $b_0 = \frac{\sqrt{3}}{2}$ as our lattice parameters.

3.2 Lattice Dynamics

3.2.1 Motivation

To attempt to describe the vortex lattice in our simulations, we will use a branch of molecular dynamics known as Langevin dynamics (LD), and was first used[19] to describe Brownian motion[20]. Brownian motion is the effect of seemingly random motion of a large particle suspended in fluid, due to the large number of collisions with the small atoms in the fluid. LD splits the effects on the large particle into two parts, a deterministic part (viscous drag), and a probabilistic effect (thermal fluctuations) which allows us to model the complex system of the particle interacting with the solvent.

If we consider the model presented by Bardeen and Stephen[14], we can understand why we are able to use LD to describe the FLL. As we have seen previously, they showed that the quasi-normal cores interact with the rest of the superconducting material, causing the viscous drag term seen in 2.19. This drag term corresponds to a solvent that can be accurately modelled by the drag term in the Langevin equation, and so we can use LD to model the FLL. We also see that the thermal fluctuations acting on the flux lines can have an effect on the vortex dynamics, as seen from the melting of the FLL and thermally activated flux creep discussed earlier. We model this as a heat bath at temperature T coupled with the vortices, and the effects of this are contained within the stochastic term in the Langevin equation, which is given by

$$m\mathbf{a} = -\gamma\mathbf{v} + \chi(t) \tag{3.1}$$

where $\mathbf{a}$ and $\mathbf{v}$ are the $3N$ components of acceleration and velocity and $\chi(t)$ is a stochastic term corresponding to the probabilistic effect of the Langevin equation. However as we discussed in section 3.1, as we are working with a thin superconductor, the vortices interact according to a 2D potential, and so we also choose that motion is confined to the 2D plane also, as the force between vortices only depends on the separation of the vortex cores. Thus, in our simulation we may treat our vortices as point particles in the 2D plane, reducing the number of components in 3.1 to $2N$.

Brandt[21] showed that for the FLL, the viscous drag is much greater than the inertia, which is equivalent to setting $m = 0$, so we see no accelerations and the vortices' motion is maximum. However from section 2.2.4, we see a repulsive force present between vortices, $\mathbf{F}_{vv}$, as well as a driving Lorentz force, $\mathbf{F}_L$, and pinning force, $\mathbf{F}_p$, on the vortices. Adding these terms to equation 3.1 gives

$$\gamma\mathbf{v} = \mathbf{F}_{vv} + \mathbf{F}_L + \mathbf{F}_p + \chi(t) \tag{3.2}$$

3.2.2 Stochastic Term

In our simulations, we have chosen the stochastic term to be described by the Andersen[22] thermostat over others, which attempts to imitate the energy fluctuations produced by the constant temperature canonical ensemble. This is done using a Poisson process designed to give thermal kicks to a particle with probability distribution $p(t) = \nu e^{-\nu t}$, where ν is the average rate of kicks. This exponential distribution is the same as a particle receiving a kick in time interval Δ with a probability $p = \nu\Delta = \Delta/\tau$ where τ is the average time between kicks. In our simulations we choose that $\nu = 10$ with $\Delta = 0.001$ for lattice melting and $\Delta = 0.005$ for driven flow, giving $p = 0.01$ and $p = 0.05$ respectively. For a Poisson process we require that the stochastic term has

$$\langle \chi_i(t) \rangle = 0 \tag{3.3}$$

$$\langle \chi_i(t)\chi_j(t') \rangle = 2k_B T\gamma\delta(t - t')\delta_{ij} \tag{3.4}$$

and from this we have the stochastic term for the i-th vortex is given by

$$\chi_i(t) = (2k_BT/\gamma p\Delta)^{\frac{1}{2}}\mathcal{N}\Theta(p - q_j) \tag{3.5}$$

where $\mathcal{N}$ is a random number chosen from a Gaussian distribution $\mathcal{N}(0,1)$, Θ is the Heaviside step function, and q_j is a random number from a random uniform distribution $\mathcal{U}(0,1)$. In our simulations, we have used the "normal_distribution" templated class, combined with default_random_engine to generate our normal distribution. To justify the use of this generator it was tested empirically, generating 100,000 random numbers, and outputting the mean and standard deviation every 50 numbers. After several tests, we found that the mean and standard deviation tend to 0 and 1 respectively, to within $< 1\%$. Similarly, with $\mathcal{U}$, we used the RNG built into the C++ random library, and generated 100,000 random numbers between 0 and 1. These numbers were sorted into bins of size 0.01, and it was found that each bin contained an equal count of numbers, correct to $< 1\%$.

3.2.3 Vortex-Vortex Interaction

The full expression for $\mathbf{F}_{vv}$ is extended from section 2.2.3 as a sum over all other vortices and is given by

$$\mathbf{F}_{vv} = \sum_{j\neq i} -\frac{\Phi_0^2}{2\pi\mu_0\lambda^3} K_1\left(\frac{|\mathbf{r}_{ij}|}{\lambda}\right)\hat{\mathbf{r}}_{ij} \tag{3.6}$$

and with no optimisations there will be N^2 calculations of the force between vortices. Reichhardt *et al.*[23] suggested that as $K_1(x)$ falls off exponentially, we can introduce a cutoff distance, $r_{cut} = 6\lambda$, beyond which we can assume the force is equal to zero, which allows us to reduce the number of calculations required. To be consistent with the work done by Besseling[2], we choose $\lambda = 1.11a_0$, and we see that $K_1(6)/K_1(1) \sim 0.0022$. In our simulations we attempted to increase this speed up further by finding a function to approximate the Bessel function, as this will be computationally quicker, but it must retain the form of the Bessel function. From work done by Koshelov and Vinokur[24], who uses the log form of the potential, we find a polynomial expression of the force $f(r) = (1/r)(1 - r^2/r_{cut}^2)^2$ (with $r_{cut} = 3.33$ as the approximation loses accuracy quickly after this). This provides a much faster (>200 times faster) expression for the force, however it overestimates the Bessel function near $r = a_0$ and underestimates for $r \gtrsim 2a_0$, as seen in figure 3.1.

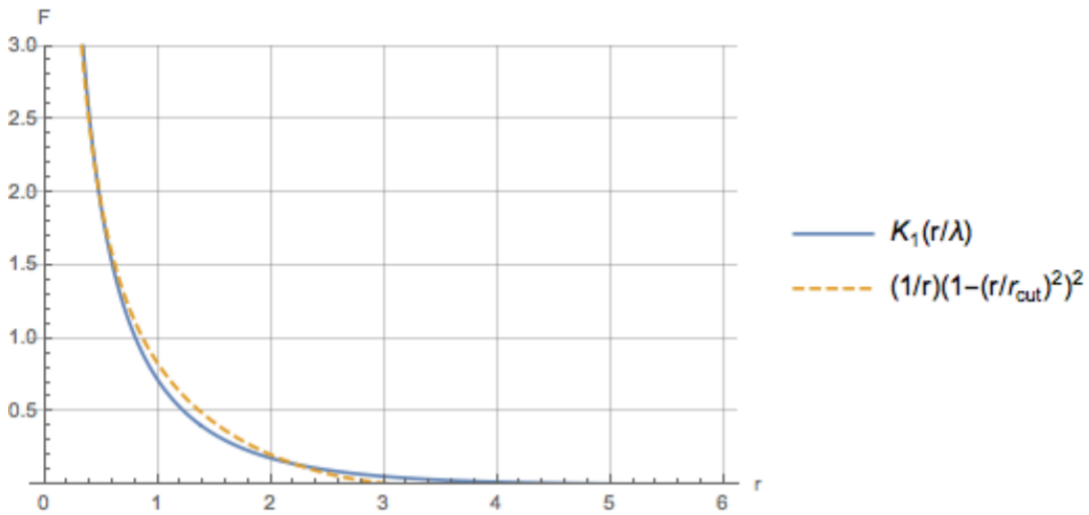

Figure 3.1: A comparison of the approximation in our simulation compared to the Bessel function. We can see that it overestimates the Bessel function near $r = a_0$ and underestimates for $r \gtrsim 2a_0$. Figure taken from Ref[1]

3.2.4 External Forces

In equation 3.2, we see that we must also consider the external forces $\mathbf{F}_L$ and $\mathbf{F}_p$, the Lorentz and pinning forces respectively. So far in our work, we have modelled several different systems, and the system we are considering determines what effect these external forces have upon the the simulation.

Firstly, when considering a pin-free system with no driving force, we omit $\mathbf{F}_L$ and $\mathbf{F}_p$ from our calculations. However, when considering flow through a single channel with boundaries of pinned vortices, these external forces must be considered to simulate this system properly. $\mathbf{F}_p$ is an attractive force that pins a vortex to a defect and decays exponentially as we move away from the defect and for this system, we initially choose the boundary vortices to be placed at pinning sites arranged in a triangular lattice. By choosing this arrangement, we can safely assume that these vortices will remain pinned to their corresponding site for the entire simulation, so instead of modelling these pinned vortices fully, we set the net force of the pinned vortices to be zero at each time step, ensuring they remain pinned. We model $\mathbf{F}_L$ as the driving force in the channel, meaning that for pinned vortices, we take $\mathbf{F}_L = 0$, and unpinned vortices $\mathbf{F}_L = F_{drive}\hat{\mathbf{x}}$, where F_{drive} is a variable that we will investigate.

3.2.5 Numerical Method

In our simulations, we used the Euler method to solve this differential equation for position, and given some initial state we can simulate the evolution of this system. We can justify the use of the Euler method over the more complex Runge-Kutta methods, of which the Euler method is the simplest, as using Runge-Kutta when we have included a stochastic term becomes much more difficult. As well as this, the Euler method has error of order $\mathcal{O}(h)$, where h is the size of the time step, and as we have used a sufficiently small time step, $\sim 10^{-3}$, our error will be sufficiently small. Due to needing a sufficiently small time step, we choose to only output the position of each vortex every 25 time steps. This means that we see a more significant evolution with each output, as well as reducing our output file size.

4 Analysing the Simulation

4.1 Basic Tests

To test our model, we can establish a few different initial states, as we know the expected outcome and this allows us to see whether our force calculations provide an accurate approximation.

4.1.1 Two Particle State

The first state we tested was very simplistic. We set up two particles in a box which is periodic in all directions, with both particles at half of the box height, one at one quarter of the width and the other at three quarters of the width. To test this state we remove the Andersen thermostat, as this test is purely to check the signs and calculations of forces due to particles considered due to the periodic boundary conditions. We expect to see that these two particles will remain stationary, as the force on each particle will be balanced due to reflected particles in the periodic boundaries. As from the end state of a simulation our forces must be balanced as the particles remain in the same position after 10,000 time steps.

4.1.2 Hexatic State

This state is a more complex version of the test in section 4.1.1. Initially, the system is set up in a perfect Abrikosov lattice, and we chose a number density of 225 vortices, before running with the Andersen Thermostat included for a large number of time steps, chosen to be 10,000. Again, we expect to see that there is little to no change in the structure of the lattice, with any changes due to large kicks from the Andersen thermostat. We can verify that our system behaves as expected, as after 10,000 timesteps, the system showed little to no disturbance. This initial state is the basis for the melting of the FLL we can produce in our simulations.

4.2 Analysis Techniques

To understand and analyse the results we produce in our simulations, we introduce a few key parameters that allow us to interpret our data.

4.2.1 Hexatic Order Parameter

Firstly, in the transition from a solid phase to a liquid phase through an intermediary hexatic phase, there is a loss of long range rotational and translational order, and we can define[3] a *Hexatic Order Parameter* for the 2D triangular lattice as a measure of the rotational order. We can also define a Translation Order Parameter, however it will not be used in our analysis and so is omitted. The Hexatic Order Parameter is defined as:

$$\Psi_H = \left| \left\langle \frac{1}{N_v} \sum_{j=1}^{N_v} \frac{1}{z_j} \sum_{k=1}^{z_j} e^{i6\theta_{jk}} \right\rangle \right| \tag{4.1}$$

where z_j, the *coordination number*, is the number of neighbours of particle j, θ_{jk} is the angle between particle j and its kth neighbour, and N_v is the number of particles. Here the angled brackets indicate a time average. Ψ_H takes a value between 0 and 1 and we expect to see that for a system initially set up in an Abrikosov lattice that for $T \ll T_m$ we have $\Psi_H \to 1$ and in the case $T \gg T_m$ we expect $\Psi_H \to 0$. Understanding how the hexatic order parameter behaves with temperature as the system evolves with time is important when analysing our data so we can clearly see under what conditions the melting transition happens. We use the Delaunay Triangulation function to calculate z_j in our analysis in Mathematica.

4.2.2 Density of Defects

We also expect that at the melting transition we should see a rapid increase in the density of dislocations, i.e. the number of vortices with a coordination number not equal to six. We define the density of defects as

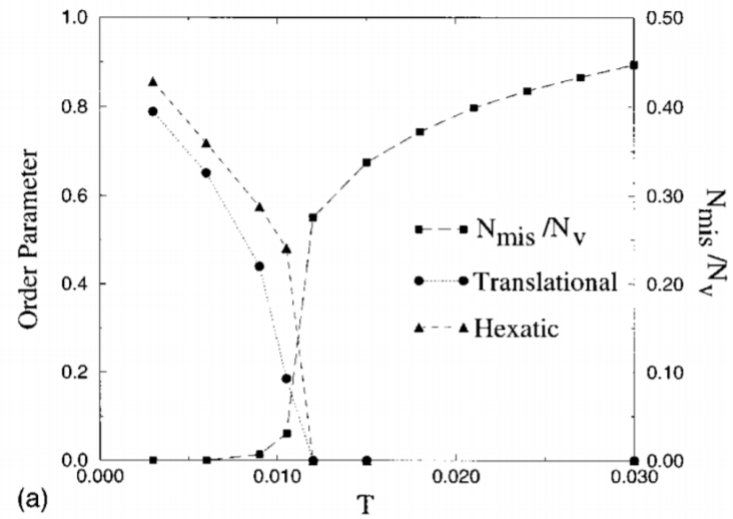

Figure 4.1: Order parameters and density of defects against temperature for the 2D periodic system of 1020 vortices. The melting transition occurs when all three parameters rapidly change, given by $T_m = 0.011$. Figure taken from Ref[3]

$$N_d = \frac{N_{dis}}{N_v} \tag{4.2}$$

where N_{dis} is the number of particles with $z_j \neq 6$, and N_v is as defined previously. If we plot Ψ_H and N_d against T we expect to see a region at which N_d rapidly increases and Ψ_H rapidly decreases, allowing us to see for which temperatures the melting transition occurs. We can see this behaviour in figure 4.1 which considers a system of 1020 vortices which we will attempt to replicate.

4.2.3 Lattice Melting

For the simulation of lattice melting we use a system of 34x30 vortices so that we can compare our results with the work by Spencer and Jensen[3], and we choose a 2D periodic system as this allows us to approximate the infinite 2D plane. This number density was chosen such that the system width L satisfies $r_{cut} < L/2$, as this condition allows us to use an optimisation to reduce the number of force calculations on each vortex.

To simulate the periodic boundary conditions without this optimisation, we had to consider image systems in the $\pm x$ and $\pm y$ directions, totalling nine regions including the original, meaning that each vortex required $9N_v - 1$ calculations of the force, where N_v is the number of vortices. However, as the optimisation condition on L is satisfied, when calculating the force on a vortex any other vortex $> L/2$ away in either the x or y direction from the original vortex can be ignored, and instead we consider the particle in the image system in the opposite direction. For example, if we consider the force on particle 1 due to particle 2 with a position $> L/2$ away in the positive x direction, the system width is subtracted from the position of the particle 2, and the force due to the image particle is now considered. This reduces the number of calculations on each vortex from $\sim 9N_v$ to $\sim N_v$, and as N_v grows with size of the system squared, we clearly see that this is a large time saving.

We choose an initial temperature $T = 0K$, and over the course of 200,000 time steps, we increase the temperature to a maximum $T = 0.02K$, with the temperature increased incrementally over 100 temperature steps, running for 10,000 time steps at each temperature. This gives a value for the temperature step of $0.001K$. This range of temperatures was chosen to attempt to give results analogous to Spencer and Jensen. We measured a value of $T_m = 0.01$ and from comparison with figure 4.1 we see our melting temperatures are approximately equal, verifying the dynamics of our simulation.

4.2.4 Critical Shear Force

Vortices that are sitting in an unpinned channel with boundaries that are pinned sit in a potential well, which from the plot of the potential in figure 4.2 it is not immediately clear. However the size of the well is very small and thus the scale for our driving force must also be, It is much more clear from the plot of the x-component of the force in figure 4.3 that vortices sit in a well, and it will also give us a scale over which to vary the driving force. So we expect that when the Lorentz force discussed in 2.2.6 is applied in the $+x$ direction, we see that below the critical value F_{Lc}, the average velocity of the vortices in the channel is $\langle v \rangle \sim 0$ but not necessarily zero due to temperature fluctuations. As the force increases past the critical value, we expect to see a rapid increase in the average velocity, and increasing driving force further the velocity should increase linearly with it.

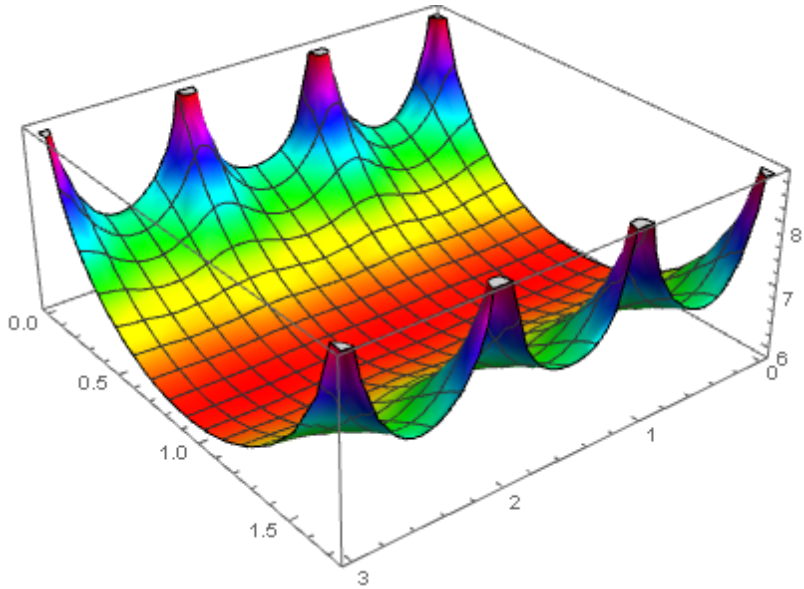

Figure 4.2: A plot of the potential across the channel over three lattice spacings in the x direction. It is not immediately clear that a potential well exists, and so we investigate the plot of the x component of the force.

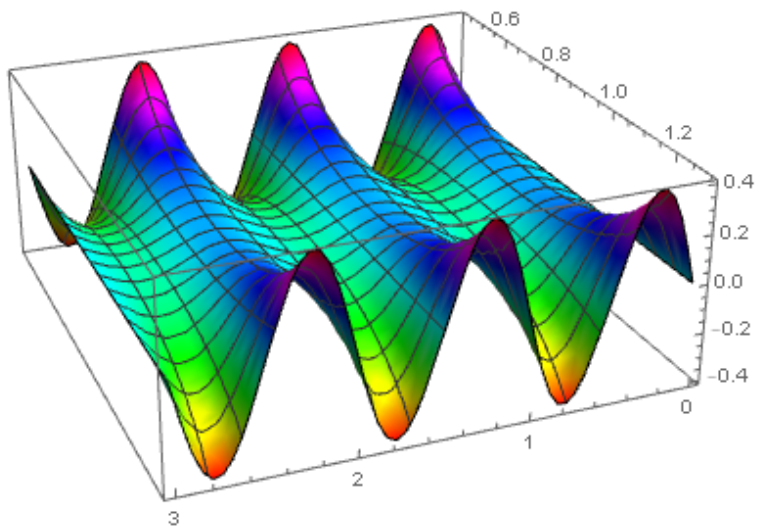

Figure 4.3: A plot of the x component of force across the channel over three lattice spacings in the x direction. It becomes much more clear that there is a potential well due to the maxima and minima that occur periodically in the x direction at $x = 0.25a_0$ and $x = 0.75a_0$

For our simulation of a commensurate single channel we used a system of 2550 vortices, to satisfy our optimisation condition as before, corresponding to 25 pinned rows on either side of a single unpinned channel. We also only require the system to be periodic in the x direction, as the channel edges are pinned,

and the force due to a vortex at $y \to \infty$ goes to zero. We choose a constant temperature of $T = 0.00001$ so that any thermal fluctuations from the stochastic term are negligible compared to the driving force and the flowing vortices remain in the channel. Based on the work by Besseling[2], we choose a maximum driving force $F_{max} = 0.1$. We then simulate $2,500,000$ time steps and increment the driving force every $50,000$ time steps, giving a value of $F_{step} = 0.002$. To measure $\langle v \rangle$, we calculate the velocity of each vortex using the distance travelled between each position output and average all of the vortices over that value of the driving force. In our model, this means we average the velocity over the 50 vortices and over $50,000$ time steps, but remembering that this only corresponds to $50,000 \div 25 = 2000$ position outputs. Due to the initial disturbance caused by the driving force in the first time step before the forces are balanced, we choose to ignore the first 50 position outputs, and so this number is revised to an average over 1950 position outputs.

From our analysis, we measure a value of the critical shear force of $F_c \approx 0.055$, seen in figure 4.4. When comparing with work from Besseling seen in figure 2.3 we measure approximately the same value for F_c, as well as observing the same shape of the curve. We can then say confidently that our simulation provides an accurate model for the system.

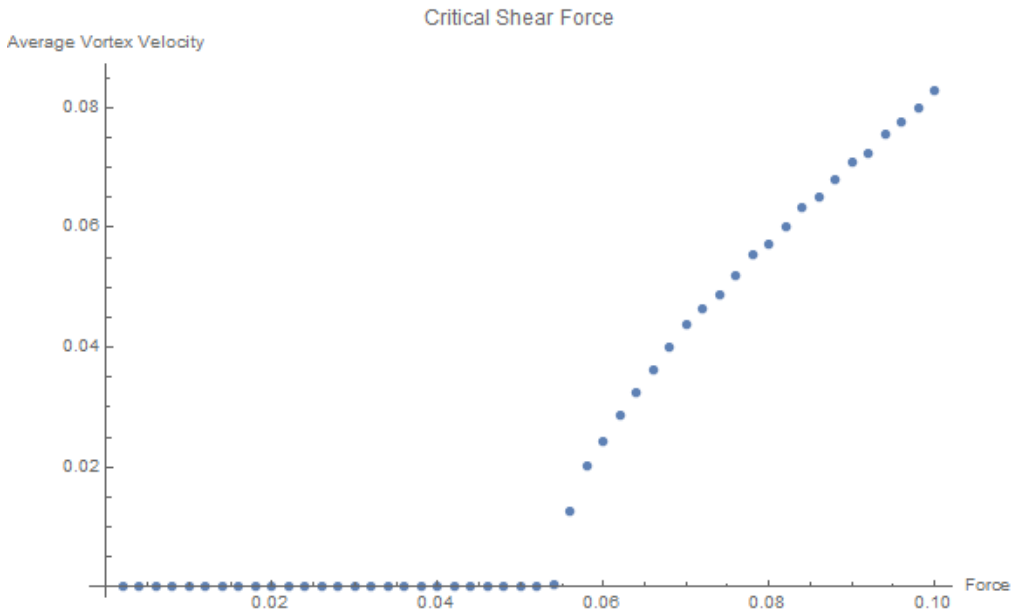

Figure 4.4: A graph of driving force against the average velocity of channel vortices, averaged over 50,000 time steps and over the 50 vortices in the channel.

5 Derivation of Analytic Form Boundary Potential

5.1 Motivation

Much of the work done when simulating the FLL is in finding ways to speed up our simulations. This can be done by optimising the way that the forces on a vortex are calculated, and most optimisations are ways to reduce the total number of calculations required for a single vortex on the lattice. However, when we are dealing with systems including a boundary, if we could find a way to derive an analytic form of this boundary, all of the calculations required when considering the force due to any vortex on a boundary could be reduced to a single calculation. As well as this, when dealing with a discrete boundary of vortices, it can be challenging to introduce changes to the structure of the boundary. When writing a simulation this can correspond to a total rewrite of the code initialising the structure, however if we have a continuous analogy to the discrete boundary, introducing changes to the structure is as simple as using a new potential.

Now that we have verified our discrete model, comparing melting temperature and critical shear force with other work[2, 3], we will attempt to find an analytic expression of the potential in the channel due to an even boundary, derived from the Bessel function form. This expression takes the form of a convergent sum, which we can then differentiate to find an expression for the force to implement into our simulation. From this point onwards, $U(x, y)$ will denote the Bessel function potential, while $V(x, y)$ will denote the analytic form of the potential. This is purely to differentiate them when discussing different simulations, however $U(x, y) \equiv V(x, y)$ remains true.

5.2 Infinite Square Lattice

We start from the potential of an infinite square lattice of vortices, extended from the expression in section 2.2.5. We set constants in front of the Bessel function equal to 1 to rescale the potential, as otherwise the values of force are very small, as well as redefining $b_0 = \sqrt{3}$. This value of b_0 is chosen so that when we construct a full triangular lattice using two square lattices, one translated by a half lattice spacing, the generated triangular lattice will have equivalent value of $b_0 = \sqrt{3}/2$. This gives:

$$V(x, y) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} K_0 \left(\frac{\sqrt{(x - na_0)^2 + (y - mb_0)^2}}{\lambda} \right) \tag{5.1}$$

which we can then apply a Fourier transform forward and back, yielding

$$\begin{aligned} V(x, y) = \sum_{n,m=-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dk_x}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{dk_y}{\sqrt{2\pi}} e^{-ik_x x - ik_y y} \\ \cdot \int_{-\infty}^{\infty} \frac{dx'}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{dy'}{\sqrt{2\pi}} e^{ik_x x' + ik_y y'} K_0 \left(\frac{\sqrt{(x - na_0)^2 + (y - mb_0)^2}}{\lambda} \right) \end{aligned} \tag{5.2}$$

Using the integral form of the Bessel function:

$$K_0(x) = \frac{1}{2} \int_0^{\infty} \frac{dt}{t} e^{\frac{-x^2}{4t} - t} \tag{5.3}$$

which is valid for $|\arg x| < \frac{\pi}{4}$, and as we are dealing with real space this will always hold. This produces

$$\begin{aligned} V(x, y) = \frac{1}{2} \sum_{n,m=-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dk_x}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{dk_y}{\sqrt{2\pi}} e^{-ik_x x - ik_y y} \\ \cdot \int_{-\infty}^{\infty} \frac{dx'}{\sqrt{2\pi}} e^{ik_x x' - \frac{(x' - na_0)^2}{4\lambda^2 t}} \int_{-\infty}^{\infty} \frac{dy'}{\sqrt{2\pi}} e^{ik_y y' - \frac{(y' - mb_0)^2}{4\lambda^2 t}} \end{aligned} \tag{5.4}$$

We can see that the integrals over x' and y' are a Gaussian form, which we can integrate by centering first, giving another calculable integral in t. This then gives:

$$\begin{aligned} V(x,y) &= \lambda^2 \sum_{n,m=-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dk_x}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{dk_y}{\sqrt{2\pi}} \frac{e^{-ik_x x + ik_x n a_0 - ik_y y + ik_y m b_0}}{1 + (\lambda k_x)^2 + (\lambda k_y)^2} \\ &= \sum_{n,m=-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dk_x}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{dk_y}{\sqrt{2\pi}} \frac{e^{-ik_x x + ik_x n a_0 - ik_y y + ik_y m b_0}}{\frac{1}{\lambda^2} + k_x^2 + k_y^2} \end{aligned} \tag{5.5}$$

We can then make use of the identity:

$$\sum_{n=-\infty}^{\infty} e^{\frac{-2\pi i k n}{T}} = T \sum_{n=-\infty}^{\infty} \delta(k - Tn) \tag{5.6}$$

Allowing us to trivially evaluate the integrals

$$V(x,y) = \frac{2\pi}{a_0 b_0} \sum_{n,m=-\infty}^{\infty} \frac{e^{-\frac{i2\pi n x}{a_0} - \frac{i2\pi m y}{b_0}}}{\frac{1}{\lambda^2} + \left(\frac{2\pi n}{a_0}\right)^2 + \left(\frac{2\pi m}{b_0}\right)} \tag{5.7}$$

This expression alone is not a good representation to use instead of the Bessel function, as it still is a sum over two variables, and it is no longer exponentially convergent. Instead we will use the semi-infinite lattice to produce an analytic form.

5.3 Semi Infinite Square Lattice

We now consider the semi-infinite lattice, infinite in the x direction and semi-infinite in y, with the particle positioned outside the lattice. We must specify the direction of y as the potential must only be periodic in x, and so we choose that y is only negative, and so the potential is described by

$$V(x,y) = \sum_{n=-\infty}^{\infty} \sum_{m=0}^{\infty} K_0 \left(\frac{\sqrt{(x - n a_0)^2 + (y - m b_0)^2}}{\lambda} \right) \tag{5.8}$$

and by following the method previously, and using equation 5.6 only in the x axis we can obtain:

$$V(x,y) = \frac{\pi}{a_0} \sum_{m=0}^{\infty} \sum_{n=-\infty}^{\infty} \frac{e^{-\frac{i2\pi n x}{a_0} - ik_y(y - m b_0)}}{\frac{1}{\lambda^2} + \left(\frac{2\pi n}{a_0}\right)^2 + k_y^2} \tag{5.9}$$

and by using the identity

$$\int_{-\infty}^{\infty} \frac{dk\ e^{-iqk}}{p^2 + k^2} = \frac{\pi}{p} e^{-|q|p} \tag{5.10}$$

we arrive at

$$V(x,y) = \frac{\pi}{a_0} \sum_{m=0}^{\infty} \sum_{n=-\infty}^{\infty} \frac{e^{-|y - m b_0| Q_n - i2n\pi x}}{Q_n} \tag{5.11}$$

with $Q_n \equiv \sqrt{\frac{1}{\lambda^2} + \left(\frac{2n\pi}{a_0}\right)^2}$. Due to our choice of y direction, and that $m \geq 0$, $b_0 > 0$ we can simplify the expression $-|y - m b_0| = -|y| - m b_0$. This allows us to perform the summation over m as we now have a geometric progression in $e^{-b_0 Q_n}$, giving an expression containing only one summation

$$V(x,y) = \frac{\pi}{a_0} \sum_{n=--\infty}^{\infty} \frac{e^{(b_0 - |y|) Q_n - i\frac{2n\pi x}{a_0}}}{Q_n \left(e^{b_0 Q_n} - 1\right)} \tag{5.12}$$

We also note that this form of the potential converges exponentially and is expressed in terms of functions that require less computation time than a Bessel function. From this semi-infinite lattice we can construct a single channel system in a triangular lattice.

5.4 Extension to Channel System

To construct the channel we add the potentials for two semi-infinite lattices, originating from each boundary and separated by the channel width L, followed by two more translated by a half lattice spacing in x and y. We express the potential as

$$\begin{aligned} V_{full} &= V(x,y) + V(x, L-y) + V(x+\frac{a_0}{2}, y+\frac{b_0}{2}) + V(x+\frac{a_0}{2}, L+\frac{b_0}{2}-y) \\ &= \frac{\pi}{a_0}\sum_{n=-\infty}^{\infty} \frac{e^{\frac{i2n\pi x}{a_0}+\frac{Q_n}{2}(b_0-L)}}{Q_n \sinh\left(\frac{b_0 Q_n}{2}\right)} \cosh\left(\left(y-\frac{L}{2}\right)Q_n\right)\left(1+(-1)^n e^{-\frac{b_0 Q_n}{2}}\right) \end{aligned} \tag{5.13}$$

which we have simplified noting our choice in coordinate system, that is, in the channel we have $0 \leq y \leq L$ and as such the argument of the modulus is always positive and the modulus can be removed. We then observe a potential well across the channel in the y direction when we set $L = b_0$. So we have now derived an analytic form of the potential in a single commensurate channel with a hexagonal lattice in the boundary, in a form that requires much less computational time. We must now differentiate to find the components of the force on a vortex in the channel.

5.5 Forces due to Analytic Potential

We can derive expressions for the components of force on a vortex in the channel using $\mathbf{F} = -\nabla V$ using our expression for the analytic potential:

$$\begin{aligned} V(x,y) &= \frac{\pi}{a_0}\sum_{n=-\infty}^{\infty} \frac{e^{\frac{i2n\pi x}{a_0}+\frac{Q_n}{2}(b_0-L)}}{Q_n \sinh\left(\frac{b_0 Q_n}{2}\right)} \cosh\left(\left(y-\frac{L}{2}\right)Q_n\right)\left(1+(-1)^n e^{-\frac{b_0 Q_n}{2}}\right) \\ &= \frac{\pi}{a_0}\left(V_0(y) + 2\sum_{n=1}^{\infty}\cos\left(\frac{2n\pi x}{a_0}\right)\frac{e^{\frac{Q_n}{2}(b_0-L)}}{Q_n \sinh\left(\frac{b_0 Q_n}{2}\right)} \cosh\left(\left(y-\frac{L}{2}\right)Q_n\right)\left(1+(-1)^n e^{-\frac{b_0 Q_n}{2}}\right)\right) \end{aligned} \tag{5.14}$$

with

$$V_0(y) = \frac{e^{\frac{Q_0}{2}(b_0-L)}}{Q_0 \sinh\left(\frac{b_0 Q_0}{2}\right)} \cosh\left(\left(y-\frac{L}{2}\right)Q_0\right)\left(1+e^{-\frac{b_0 Q_0}{2}}\right) \tag{5.15}$$

5.5.1 F_x

Differentiate with respect to x to give:

$$F_x(x,y) = -\frac{\partial V}{\partial x} = \frac{4\pi^2}{a_0^2}\sum_{n=1}^{\infty} n\sin\left(\frac{2n\pi x}{a_0}\right)\frac{e^{\frac{Q_n}{2}(b_0-L)}}{Q_n \sinh\left(\frac{b_0 Q_n}{2}\right)} \cosh\left(\left(y-\frac{L}{2}\right)Q_n\right)\left(1+(-1)^n e^{-\frac{b_0 Q_n}{2}}\right) \tag{5.16}$$

We see that the value of the x component of the force is periodic in x with period a_0, having maxima and minima at $a_0/4$ and $3a_0/4$.

5.5.2 F_y

Differentiate with respect to y to give:

$$F_y(x,y) = -\frac{\partial V}{\partial y} = \frac{\pi}{a_0}\left(-\frac{\partial V_0}{\partial y} + \sum_{n=1}^{\infty}\cos\left(\frac{2n\pi x}{a_0}\right)\frac{e^{\frac{Q_n}{2}(b_0-L)}}{\sinh\left(\frac{b_0 Q_n}{2}\right)} \sinh\left(\left(y-\frac{L}{2}\right)Q_n\right)\left(1+(-1)^n e^{-\frac{b_0 Q_n}{2}}\right)\right) \tag{5.17}$$

with

$$\frac{\partial V_0}{\partial y} = \frac{e^{\frac{Q_0}{2}(b_0-L)}}{\sinh\left(\frac{b_0 Q_0}{2}\right)} \sinh\left(\left(y-\frac{L}{2}\right)Q_n\right)\left(1+e^{-\frac{b_0 Q_0}{2}}\right) \tag{5.18}$$

We observe a well in the potential at $y = b_0/2$, where the value of $F_y = 0$. This value of y is significant as it corresponds to the centre of a single channel.

5.5.3 Channel Vortex Interactions

It is important to note that when we simulate our model, these expressions for the force on a vortex are only the contributions from vortices in the boundary. We must still model the vortex-vortex interactions between the free vortices in the channel using the same method as the discrete case, with a value of $r_{cut} = 3\lambda$.

6 Verification of the Analytic Potential

6.1 Convergence

Firstly, we will begin by analysing the convergence of the analytic potential compared to the Bessel function with respect to the length of the sum, N. As both forms of the potential are symmetric on $y = 0.5b_0$ and periodic in x, we will only need to analyse the convergence in $y = (0.5b_0, b_0)$ and over one x period. We will compare the value of these functions to the "true" value, determined using the potential due to the Bessel function, constructed similarly to section 6.4 for the analytic form, and summed over 100 vortices in either direction, given by:

$$U_{full}(x, y) = U(x, y) + U(x, L - y) + U(x + \frac{a_0}{2}, y + \frac{b_0}{2}) + U(x + \frac{a_0}{2}, L + \frac{b_0}{2} - y) \tag{6.1}$$

where

$$U(x, y) = \sum_{n=-100}^{N=100} \sum_{m=-100}^{0} K_0 \left(\frac{\sqrt{(x - na_0)^2 + (y - mb_0)^2}}{\lambda} \right) \tag{6.2}$$

We chose this length of sum as the expression in equation 6.1 converges for $N < 30$, meaning that if we choose $N = 100$ we expect that this will be an accurate value of the potential at any point. As well as this, by inspecting the analytic function for $0.01b_0 < y < 0.99b_0$, we found that the analytic function would converges for $N < 100$, and so the choice of $N = 100$ for the true value ensures that the ratio of the analytic function to the true value remains relevant also. Beyond these values of y the analytic form begins to break down, however we do not expect to find any vortices outside of this region when starting in the ground state due to the repulsive nature of the force from the boundary.

6.1.1 Convergence along x-axis

In our simulations, we work in the centre of the channel, $y = 0.5b_0$, and the maximum and minimum values of the force due to the boundaries are found at $x = 0.25a_0$ and $x = 0.75a_0$. We will therefore analyse the convergence of the analytic potential at these critical points and also in the ground state position at $x = 0.5a_0$. This will allow us to verify that the analytic potential is valid, as well as allow us to determine the optimum value of N to use in the driven simulation by finding a balance between accuracy and speed.

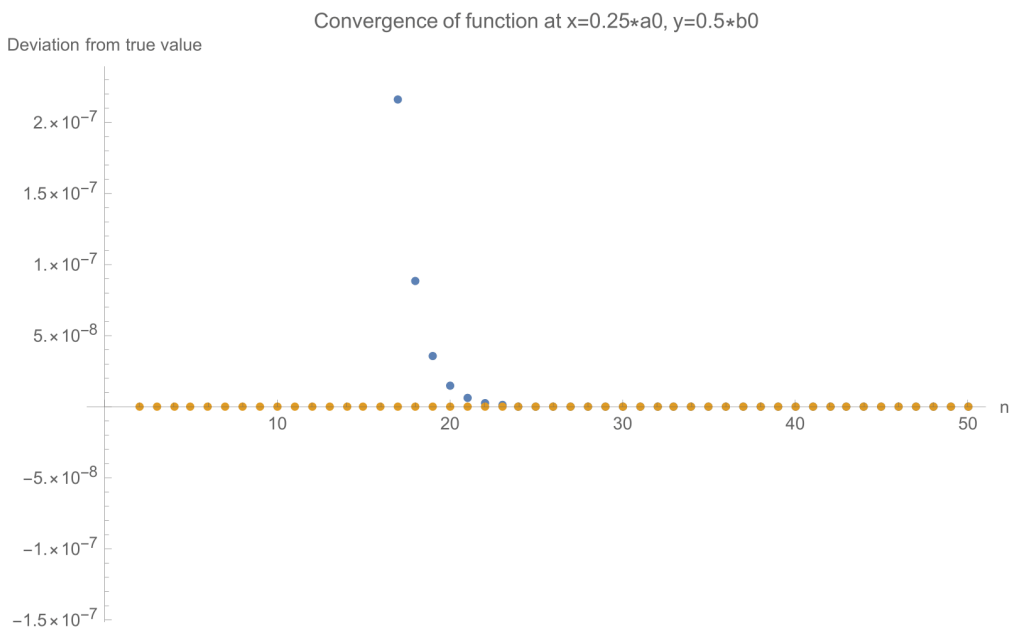

Figure 6.1: Convergence of the analytic potential and the Bessel potential at the point $(0.25a_0, 0.5b_0)$ at varying sum lengths. y-axis given by $1 - \frac{V}{V_{true}}$.

We can clearly see from figs. 6.1 to 6.3 that as long as a vortex remains in the centre of the channel, the analytic solution appears to converge completely within $N = 3$. Due to the scale of the figures we cannot

see the deviation from the true value for $N = 1$, but it is calculated as $\approx 3 \times 10^{-6}$, which is significantly small enough that we may justify using $N = 1$ when modelling our simulations using the analytic function at these coordinates.

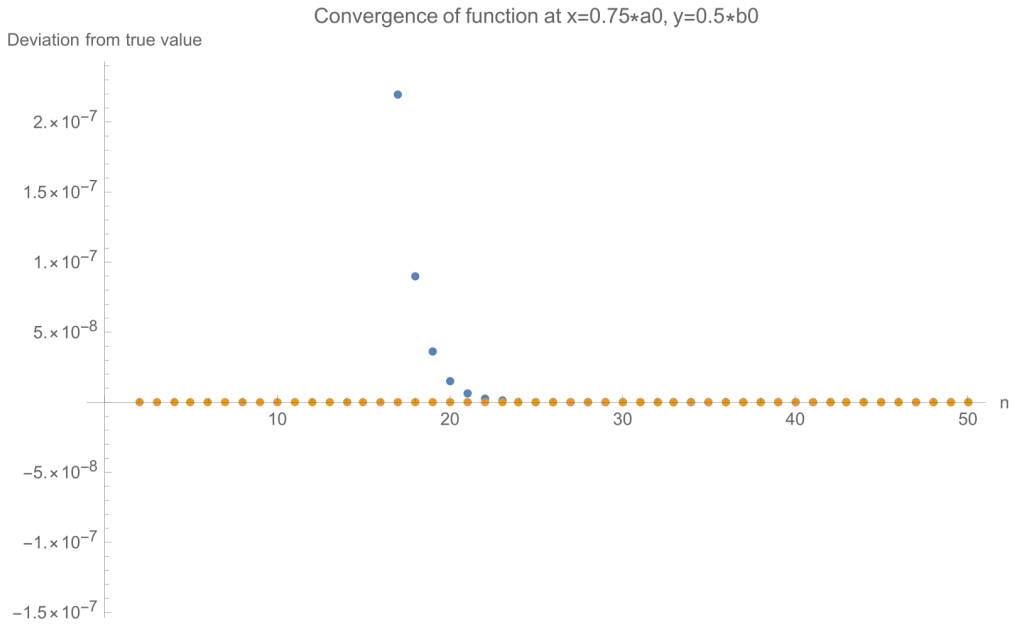

Figure 6.2: Convergence of the analytic function and the Bessel potential at the point $(0.75a_0, 0.5b_0)$ at varying sum lengths. y-axis given by $1 - \frac{V}{V_{true}}$.

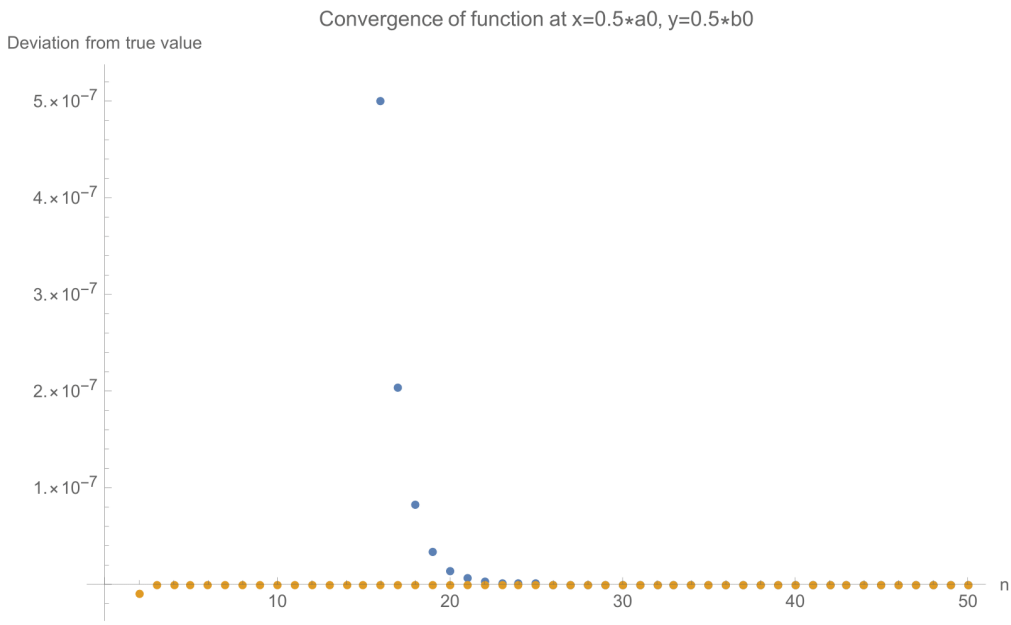

Figure 6.3: Convergence of the analytic function and the Bessel potential at the point $(0.5a_0, 0.5b_0)$ at varying sum lengths. y-axis given by $1 - \frac{V}{V_{true}}$.

6.1.2 Convergence along y-axis

The convergence of the analytic function in the y-direction is less significant to our simulations, as we expect a vortex at $y = 0.5b_0$ to remain so when considering driven flow in a single commensurate channel. However by analysing the convergence, if we are given a value of N we may determine a domain of convergence around the vortex in which the analytic form remains accurate. Alternatively, given a radius around the vortex we desire to work in, we can choose a value of N to ensure that the analytic form fully converges. In figure 6.5 we see that at a value of $y = 0.949b_0$, both the analytic form and the Bessel form

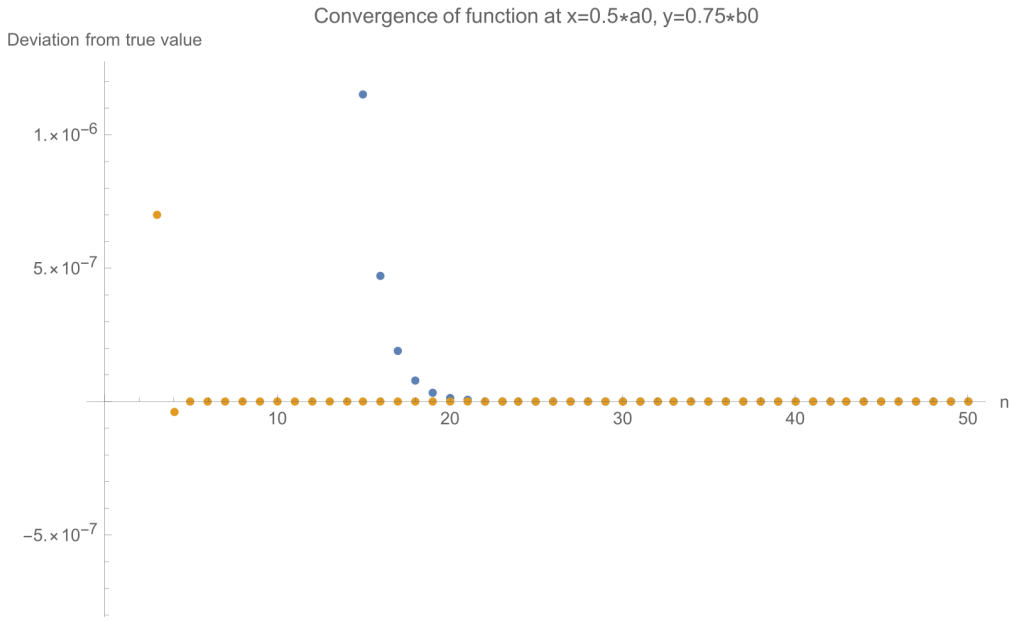

Figure 6.4: Convergence of the analytic function and the Bessel potential at the point $(0.5a_0, 0.75b_0)$ at varying sum lengths. y-axis given by $1 - \frac{V}{V_{true}}$.

potentials converge for similar values of N. We should note here that the Bessel form is a sum over square space, so whilst the value of N is similar the Bessel form will still be slower, even before considering the computation time of one vortex contribution from the Bessel function against one term in the analytic sum.

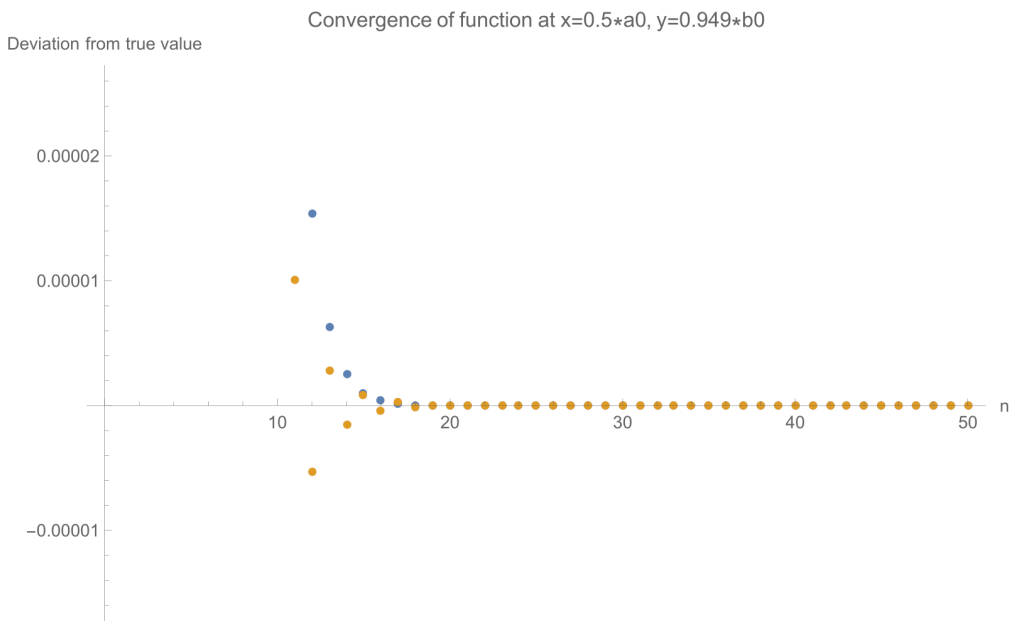

Figure 6.5: Convergence of the analytic function and the Bessel potential at the point $(0.5a_0, 0.949b_0)$ at varying sum lengths. y-axis given by $1 - \frac{V}{V_{true}}$.

6.1.3 Conclusions

We shall define the domain in which the analytic form potential is more appropriate to use as $\mathbf{D} = (x, y) = \mathbb{R} \times (0.05b_0, 0.95b_0)$, as the analytic form converges faster than the Bessel form in this region. Provided we can then show that vortices starting in $\mathbf{D}$ will remain in $\mathbf{D}$, we may use only the analytic form potential to calculate the force on a vortex in the channel due to the boundary. From our data, we can also conclude that for our simulations of driven flow we can choose our sum in equation 5.14 at

leading order. These conclusions are justified because the value of the y component of force due to vortex interactions at $y = 0.5b_0$ is zero for all x. As we have no external forces in the y-direction and choose for all vortices to begin at $y = 0.5b_0$, we know that the vortices will always remain at $y = 0.5b_0$, therefore satisfying the relevant conditions.

6.2 Critical Shear Force

6.2.1 Results

In figure 6.6 we can see our analysis of flow of 50 vortices in a single commensurate channel using an analytic form potential to describe the boundaries.

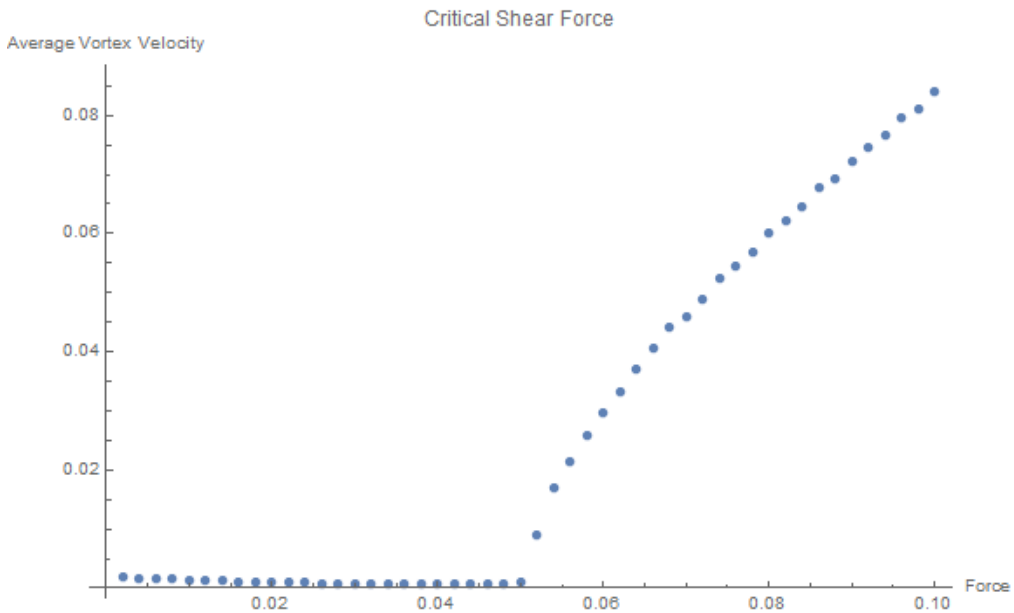

Figure 6.6: A graph of driving force against average velocity of channel vortices when the boundary is described by an analytic form potential. The velocity is averaged over 50,000 time steps and over the 50 vortices in the channel.

We measure a value of the critical shear force $F_c \approx 0.05$ for our analytic form potential. The discrepancy between this value and the value determined in section 5.2.4 can be explained due to our choice of the value of r_{cut}. As we have seen in section 7.1, the Bessel form of the potential converges fully when considering the sum over $N = 25$, corresponding to vortices at a maximum distance of $r = \sqrt{(25a_0)^2 + (25b_0)^2} = 50$ away for out values of a_0 and b_0. As we only consider vortices within $r_{cut} = 3\lambda$ in our discrete case due to the force approximation, it turns out the value of $F_c \approx 0.55$ measured previously is higher than the true value, which we have observed in the analytic case. If we considered the discrete case using the Bessel function instead of an approximation, and considered vortices with $r_{cut} = 50$, we should expect to see a value of F_c similar to the one we have measured for the analytic potential. To do this is not entirely feasible, as the simulation time will be far too long, given that our discrete case simulation with several optimisations and $r_{cut} = 3\lambda$ took over 45 minutes to complete.

6.3 Computation Times

One of the key motivations behind this project has been to find a way to decrease simulation times, as this remains one of the main obstacles for research investigating vortex matter. Because the analytic potential consists of elementary functions, and we only consider up to $|n| \leq 1$ for the simulation of driven flow, we expect that calculation of force on a vortex will be much faster than a Bessel function. As well as this, the analytic form describes the entire channel edge in one function, so only one calculation is required for each vortex compared to $\sim S^2$ in the discrete case, where S is the system length. This means that we should expect to see a drastic time save even when comparing to the simulation modelled using a force approximation.

For comparison, the simulation in section 4.2.4 ran for a total time of $\approx 2700s = 45$ minutes. Even when we considered the sum in the analytic form up to $n = 10$ we found that it took a total time of $\approx 750s = 12.5$ minutes. Further speed up occurred when reducing to $n = 1$ as discussed in section 6.1.3, with the simulation only requiring $\approx 475s \approx 8$ minutes. We can clearly see that our objective of reducing simulation time has been fulfilled, as the length of the simulation now only depends on the length of sum chosen and with the number of vortices in the channel.

7 Future Work

Through our derivation of the analytic potential, we have an expression for the potential in terms of elementary functions with the x dependency coming from a cosine for $|n| \leq 1$. We are now able to investigate the formation and behaviour of dislocations in the lattice structure when altering the lattice structure in some way, either by a modulation in the boundary or changing the lattice parameters. For example, in previous work [2, 23] on commensurate and incommensurate channel systems, a channel that is incommensurate with the row spacing causes plastic flow that is mediated by dislocations that form in the lattice structure.

7.1 Modulation of the Analytic Potential

One method of altering the structure of the lattice involves a modulation of the analytic potential. We take the potential in equation 5.14 at leading order and apply a modulation to the cosine in the x direction, of the form.

$$\cos(x) \longrightarrow \cos(x)\left(1 + \frac{A}{2}e^{-\frac{(x-x_0)^2}{\sigma^2}}\right)$$

where x_0 is the centre of the modulation, σ is the spread of the modulation, and A is its amplitude. A modulation of this kind corresponds to an increase or decrease in density of vortices in the channel edge, depending on the sign of A. Taking this new potential, we again are only required to differentiate to find the components of force. It becomes obvious that it is much simpler to implement a new expression for the force than to write a new program in the discrete case that contains the variation in channel edge density.

7.2 Altering Lattice Geometry

Another way in which we can alter our systems to investigate the behaviour of dislocations is by altering the geometry of our lattice. For example, simply by changing the value of L to a non-integer value of b_0 would produce an incommensurate channel system. Through previous work [2] we can expect to see plastic flow due to the dislocations that form. We could then use this work to compare the behaviour of our analytic form potential and validate that it behaves as expected.

We can extend our system to include multiple rows in the channel by increasing L and modelling multiple rows discretely in the channel, however the form we obtained is only for odd numbers of rows in the channel. To construct the system of an even number of channel we set our value of $L = pb_0/2$ where $p = 1, 3, 5...$ and as such we have

$$V(x,y) = \frac{\pi}{a_0}\sum_{n=-\infty}^{\infty}\frac{e^{\frac{i2n\pi x}{a_0}+\frac{Q_n}{2}(\frac{b_0}{2}-L)}}{Q_n\sinh\left(\frac{b_0 Q_n}{2}\right)}\cosh\left(\left(y-\frac{b_0}{4}-\frac{L}{2}\right)Q_n\right) + (-1)^n\cosh\left(\left(y-\frac{b_0}{4}-\frac{L}{2}\right)Q_n\right)$$

and using this form allows us to construct channel systems with even numbers of rows.

8 Conclusion

In this project, we have performed simulations and analysis of vortex matter in a type II superconductor, with specific focus on the system of driven flow in a single channel with pinned channel edges. We first constructed a model to describe the interactions between the vortices in a 2D periodic space using an approximation to the Bessel function that governs vortex-vortex interactions. We verified that this approximation was accurate enough by melting our lattice, and comparing the results to work by Spencer and Jensen[3].

We then moved to constructing a single commensurate channel system with pinned channel edges using this force approximation. By applying a driving force in the $+x$ direction we were able to measure a critical shear force, the value of the driving force at which the unpinned vortices in the channel begin to flow, measured as $F_c \approx 0.055$. This result was verified against work by Besseling *et al.*[2].

The main aim of this project was to find an analytic form of the potential due to the pinned channel edges, derived from the integral form of the Bessel function. Through applying Fourier transforms and various identities, we arrived at an expression for the potential in a channel, with an odd number of rows, due to the channel edges, given by

$$V(x,y) = \frac{\pi}{a_0} \sum_{n=-\infty}^{\infty} \frac{e^{\frac{i2n\pi x}{a_0} + \frac{Q_n}{2}(b_0 - L)}}{Q_n \sinh\left(\frac{b_0 Q_n}{2}\right)} \cosh\left(\left(y - \frac{L}{2}\right) Q_n\right) \left(1 + (-1)^n e^{-\frac{b_0 Q_n}{2}}\right)$$

with

$$Q_n \equiv \sqrt{\frac{1}{\lambda^2} + \left(\frac{2n\pi}{a_0}\right)^2}$$

This form of the potential is expressed in terms of elementary functions, and describes the contribution from all of the vortices in the channel. We were also able to show that in the channel centre, we are only required to consider the sum over $|n| \leq 1$, as the potential converges to within $\approx 3 \times 10^{-6}$ of the true value.

Using this form of the potential, we again constructed a single commensurate channel system. Applying a driving force, we measured a value of the critical shear force of $F_c = 0.05$. Comparing against the value measured in the discrete case, we see there is a discrepancy which can be explained by the existence of a cut off distance optimisation in the discrete simulation. Due to the value of $r_{cut} = 3\lambda$, chosen so the force approximation remained accurate, vortices outside this range were not considered despite still having a measurable effect. We suspect that if we used the Bessel function form and used $r_{cut} = 20\lambda$, we might expect to see a lower value of F_c in the discrete case. However, the discrete simulation already required long computation times even with various optimisations, so performing this simulation may not be entirely feasible.

We measured a significant decrease in the computation time required to perform the simulation when using the analytic potential description in comparison to the discrete case. The total simulation time for the discrete case was ≈ 2750s, while for the analytic potential case had a total running time of ≈ 475s. This speed up is as the force due to the boundary on a single vortex is independent of system size.

As such we have definitely achieved our aim of producing an analytic form that describes the channel edges with the purpose of reducing simulation times. In future we may wish to perform the discrete form simulation we described with $r_{cut} = 20\lambda$, as we expect to measure a value of F_c that matches more closely with the analytic form. This would allow us to completely justify the use of the analytic potential as a description of the boundary in a channel system.

References

[1] Jonathan Watkins. *Density Driven Vortex Motion in Narrow Channel Superconductors.* PhD thesis, University of Birmingham, 2015.

[2] R Besseling, P H Kes, T Dröse, and V M Vinokur. Depinning and dynamics of vortices confined in mesoscopic flow channels. *New Journal of Physics*, 7:71–71, 2005.

[3] Steven Spencer and Henrik Jeldtoft Jensen. Absence of translational ordering in driven vortex lattices. *Physical Review B*, 55(13):8473–8481, 1997.

[4] H. Kamerlingh Onnes. Further experiments with liquid helium. *Leiden Comm.*, 1911.

[5] W. Meissner and R. Ochsenfeld. Ein neuer Effekt bei Eintritt der Supraleitfähigkeit. *Die Naturwissenschaften*, 21(44):787–788, 1933.

[6] Tinkham. *Introduction to Superconductivity.* McGraw Hill, 2 edition, 1996.

[7] F London and H London. The electromagnetic equations of the supraconductor. *Proceedings of the Royal Society of London. Series A - Mathematical and Physical Sciences*, 149(866):71–88, 1935.

[8] V. L. Ginzburg and L. D. Landau. *Zh. Eksperim. i Teor. Fiz.*, 20:1064, 1950.

[9] Adwaele. Image from wikipedia, 2012.

[10] T. N. P. Foundation. The nobel prize in physics 2003. http://www.nobelprize.org/nobel_prizes/physics/laureates/2003/abrikosov-bio.html.

[11] J. Bardeen, L. N. Cooper, and J. R. Schrieffer. Theory of superconductivity. *Physical Review*, 108(5):1175–1204, 1957.

[12] A. P. Drozdov, M. I. Eremets, I. A. Troyan, V. Ksenofontov, and S. I. Shylin. Conventional superconductivity at 203 kelvin at high pressures in the sulfur hydride system. *Nature*, 525(7567):73–76, 2015.

[13] A. A. Abrikosov. On the magnetic properties of superconductors of the second group. *Soviet Physics JETP-USSR*, 5:1174–1183, 1957.

[14] John Bardeen and M. J. Stephen. Theory of the motion of vortices in superconductors. *Physical Review*, 140(4A):A1197–A1207, 1965.

[15] David R. Nelson and B. I. Halperin. Dislocation-mediated melting in two dimensions. *Physical Review B*, 19(5):2457–2484, 1979.

[16] M.Cristina Marchetti and David R. Nelson. Dynamics of flux-line liquids in high-tc superconductors. *Physica C: Superconductivity*, 174(1-3):40–62, 1991.

[17] Q-H Chen, Q-M Nie, J-P Lv, and T-C Au Yeung. Elastic theory for the vortex-lattice melting in iron-based high-tc superconductors. *New Journal of Physics*, 11(3):035003, 2009.

[18] Vílchez. Image from wikipedia, 2008.

[19] Don S. Lemons, Anthony Gythiel, and Paul Langevin. Paul Langevin's 1908 paper "on the theory of brownian motion" ["sur la théorie du mouvement brownien," C. R. Acad. Sci. (Paris) 146, 530–533 (1908)]. *American Journal of Physics*, 65(11):1079–1081, 1997.

[20] Albert Einstein. The motion of elements suspended in static liquids as claimed in the molecular kinetic theory of heat. *Annalen der Physik*, 17:549–560, 1905.

[21] E H Brandt. The flux-line lattice in superconductors. *Reports on Progress in Physics*, 58(11):1465–1594, 1995.

[22] Hans C. Andersen. Molecular dynamics simulations at constant pressure and/or temperature. *The Journal of Chemical Physics*, 72(4):2384–2393, 1980.

[23] C. Reichhardt, C. J. Olson, R. T. Scalettar, and G. T. Zimányi. Commensurate and incommensurate vortex lattice melting in periodic pinning arrays. *Physical Review B*, 64(14), 2001.

[24] A. E. Koshelev and V. M. Vinokur. Dynamic melting of the vortex lattice. *Physical Review Letters*, 73(26):3580–3583, 1994.